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689 employees, 15 features

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Problem Statement

The Problem

Organizations manually set salaries using guesswork — leading to inconsistency, budget overruns, and employee dissatisfaction across departments and countries.

The Goal

Build a data-driven ML model that predicts employee annual salary using measurable factors, and deploy it as a real-time web application for HR decision-making.

Predict Salary

Accurate annual salary estimate

Standardize Pay

Consistent decisions across teams

Deploy Live

Interactive web app for real use

Dataset Overview

689

Employees

15

Features

5

Countries

0

Missing Values

Feature	Description	Role
Years	Years at company	Input ✓
Job Rate	Performance score 1–5	Input ✓
Annual Salary	Yearly compensation (USD)	🎯 Target
Department	Business unit	EDA only
Country	Egypt, KSA, UAE, Syria, Lebanon	EDA only
Overtime Hours	Total OT worked	EDA only

Exploratory Data Analysis — Key Insights

Gender Split

65% Male / 35% Female

Workforce is predominantly male — worth monitoring for equity.

Job Rate

Mean: 3.59 | Clusters at 3 & 5

Bimodal distribution — most employees score average or high.

Overtime Hours

Median: 7 hrs | Max: 198 hrs

Heavily skewed — a few employees work extreme overtime hours.

Annual Salary

\$8,436 – \$41,400

\$33K spread across roles, departments, and geographies.

By Department

Significant variation

Top departments earn noticeably more than lower-tier ones.

By Country

5 countries, varying rates

Egypt, KSA, UAE, Syria, Lebanon show distinct job rate patterns.

Machine Learning Model

Algorithm: Linear Regression

$$\text{Annual Salary} = \beta_0 + \beta_1(\text{Years}) + \beta_2(\text{Job Rate})$$

- Supervised learning for continuous target
- Interpretable and fast to train
- Strong baseline model

Train / Test Split

TRAIN — 551 (80%)

TEST — 138 (20%)

sklearn train_test_split

Model Results

Mean Absolute Error (MAE): \$7,703

Salary range spans \$33K — MAE is a reasonable baseline for a 2-feature model.

Adding Department, Country, and Overtime Hours is expected to reduce error significantly.

Streamlit Web Application

Salary Prediction App

This app predicts the salary of an employee based on Years and Job Rate

Enter the Year

Enter the Job Rate

Press the button for salary prediction

App Features

Year Input

Numeric input for years of experience

★ Job Rate Input

0.0 – 5.0 in 0.5 increments

One-click Predict

Instant salary prediction on button press

Result Display

Formatted output with balloon animation

Saved Model

Loads linearmodel.pkl via joblib

Next Steps & Improvements

Add More Features

Include Department, Country, Overtime Hours, and Center using One-Hot Encoding to significantly reduce MAE.

Try Better Models

Test Random Forest, XGBoost, and Gradient Boosting to capture non-linear salary relationships.

Cross-Validation

Replace single train/test split with k-fold cross-validation for more reliable performance estimates.

Deploy Online